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14 March 1986
SUPERSEDING
MIL-T-52698
31 August 1970

MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, TIERING, WALKIE,

STRADDLE TYPE, ELECTRIC, 2,000- AND 2,500-POUNDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric walkie straddle trucks for use in warehouses, the load capacity of which is rated at 24 inches from the heel of the forks.

1.2 Classification. Trucks shall be one of the following sizes as specified:

Type E, size 1 - 2,000 pounds capacity, 130 inches minimum lift height, 83 inches maximum collapsed mast height.
Type E, size 2 - 2,500 pounds capacity, 130 inches minimum lift height, 83 inches maximum collapsed mast height.
Type EE, size 3 - 2,000 pounds capacity, 130 inches minimum lift height, 83 inches maximum collapsed mast height.
Type EE, size 4 - 2,500 pounds capacity, 130 inches minimum lift height, 83 inches maximum collapsed mast height.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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SPECIFICATIONS

FEDERAL

W-B-133	- Battery Storage (Lead Acid, Industrial Portable Service).
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-268	- Test and Inspection of Trucks, Lift, Fork.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DOL)

29CFR1910 - Occupational Safety and Health Administration.

(Application for copies should be addressed to a local DOL officer or the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the

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issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B56.1 - Safety Standard For Powered Industrial Trucks.
MH11.4 - Forks and Fork Carriers for Powered Industrial Trucks.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX.
Welder Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

AMERICAN WELDING SOCIETY (AWS)

Welding Procedure and Performance Qualification, AWS B2.1.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook - Recommended Practices.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).

TIRE AND RIM ASSOCIATION, INC. (TRA)

Tire and Rim Association Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OR 44313.)

UNDERWRITERS LABORATORIES, INC.

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ANSI/UL-583 - Standards for Safety (Power-Operated Industrial Trucks).

(Application for copies should be addressed to Underwriters Laboratories, Inc., 161 Avenue of the Americas, New York, NY 10013; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Lane, Santa Clara, CA 95050.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Guide to NEMA Standards Publication - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The walkie straddle truck (hereinafter referred to as "truck") shall be driven by an electric motor with an industrial battery as a power source. It shall consist of: two straddle legs, a tiering mast, and a manual T-shaped steering handle with controls incorporated on the handle.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of trucks from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

3.2.1 Certificates of compliance. When specified (see 6.2), or upon request, a certificate of compliance with the requirements of this specification shall be furnished. Back-up data to support the certificate of compliance may be requested by the contracting officer for review. The certificate of compliance shall read as follows:

This certifies that the _____(component, part or assembly) complies with all provisions and requirements of specification _____ paragraph_____. The certificates shall be signed by a duly authorized agent of the contractor.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Castings. Castings shall be of the quality normally used in the materials handling industry for this type of truck, and the castings shall be free from blowholes, porosity, shrinkage, cracks, or other defects. Castings shall not be repaired to adversely affect the strength, or intended use. Sufficient finish shall be allowed to permit machining for leak -free sealing at covers, closure points, and fasteners.

3.3.5 Bearings. Unless otherwise specified herein, rotating parts shall be mounted on ball or roller bearings to the extent normally provided in the material handling industry for this type of truck. Bearing load capacity, including applicable safety and life factors, as recommended by the gearing manufacturer, shall be equal to or greater than maximum bearing loading required for the truck to complete all operations specified herein.

3.3.6 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. The types of lubricants shall be selected from, and be in accordance with, SAE J754. All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Grease lube fittings shall conform to SAE J534.

3.3.7 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the trucks may be newly fabricated from recovered materials to the maximum extent practicable, provided the trucks produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the trucks.

3.4 Construction. Trucks shall be constructed for the rated load, at a 24-inch load center measured from the face of the forks, and at a minimum lift height of 130 inches. Truck construction shall insure safe, efficient, and economical operation under all operating conditions specified herein. There shall be no evidence of accelerated wear, failure, or permanent deformation when the trucks are operated under any condition specified herein.

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3.4.1 Safety. The truck shall conform to the applicable requirements of ANSI B56.1 and ANSI/UL583 for type E and type EE trucks and such additional requirements as specified herein, with the Occupational Safety and Health Standards requirements of 29 CFR, chapter XVII, part 1910 and all pertinent national safety standards in effect at the time of solicitation. Load backrest extension and overhead guards are applicable.

3.5 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system components or assembly where applicable, battery, wearing parts of lift and tilt mechanism, brakes and components, wear parts of steering assembly, tires, wheels, lights, and horn, without the removal of any major parts other than covers. If hand access openings are used on the exterior of the truck, the edges of each opening shall be smooth and the opening shall be provided with a removable or hinged cover, whenever it is required. Dimensions of access openings shall be in accordance with SAE J925 for bare hand. There shall be no interference with the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions. Drain holes with removable drain plugs shall provide for complete drainage without disconnection of hoses.

3.6 Electrical equipment. All electrical equipment and wiring incorporated in the trucks shall conform to the applicable requirements of ANSI/UL-583, unless otherwise specified herein.

3.6.1 Overcurrent devices. In addition to the requirements of 3.4.1, overcurrent devices shall be provided in each power circuit and each control circuit to protect the circuit from short circuits and prolonged overloads.

3.6.2 Control panel. Construction of the control panel shall permit removal of the panel as a complete unit, and for removal of assemblies and components mounted thereon. All assemblies and components shall be mounted on the control panel with permanently secured bolts, studs or nuts. These bolts, studs and nuts shall not turn or become loosened when an assembly or component is removed or replaced. The panel cover shall be free of any obstruction or interference from the truck structure which would prevent removal of the cover for inspection and servicing. All control components shall operate on 12 volts dc or 24 volts dc. Contactor tips shall be of a copper or silver alloy composition, and shall be replaceable using ordinary hand tools.

3.6.3 Conductors. Conductors shall be correctly sized, color coded, and insulated in accordance with applicable standards of the Underwriters Laboratories ANSI/UL-583, and the National Electrical Manufacturers Association.

3.7 Motors and motor accessories.

3.7.1 Motors. Motors shall be heavy-duty industrial type with high starting torque. Motors shall not fail when the truck is tested as specified in 4.3.2. Insulation for motors shall be class B, F, or H as defined in NEMA Standards. Motor protection shall be as specified in ANSI/UL 583 for trucks type E or EE.

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3.7.1.1 Speed controller. The speed controller shall provide not less than two forward and two reverse speeds. The controller shall: (1) provide means to permit precision maneuvering in restricted areas, (2) provide smooth acceleration between each successive speed range, and (3) have automatic electrical or mechanical interlocks to prevent starting or reversing except through forward or reverse first speed, respectively. There shall be maximum interchangeability of parts between contractors on a given truck. Contractors shall operate on 12 volts dc or 24 volts dc and shall be replaceable. There shall be no evidence of failure in any part of the controller when the truck is tested as specified in 4.4. When specified, the speed controller shall be the solid state type and shall provide stepless speed control in both forward and reverse directions.

3.7.2 Battery. The truck shall be furnished with industrial type battery(s) having a capacity of not less than 432 AH at 6-hour rate at 12 volts or 215 AH at the 6-hour rate at 24 volts. Battery shall meet the requirements of W-B-133.

3.7.3 Drive unit. All rotating parts shall be supported on anti-friction bearings. The drive shall be a gear reduction unit operating in a constant oil bath. The drive gear housing shall be equipped with a magnetic drain plug and a filler plug. The motor reduction unit and drive wheel assembly shall be integrally mounted and shall revolve on antifriction bearings. The drive motor shall not fail when the truck is tested as specified in 4.3.2.

3.7.4 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the forks and mast. Hydraulic fittings on pumps, valves, and cylinders shall be the O-ring straight thread type as specified in SAE J514. The system shall provide for lowering of rated load at the rate specified in 3.10.3. A pipe tap or straight thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. All pressure hose fittings shall be 37-degree flare type female in accordance with SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum relief valve setting. Pressure tube fittings shall conform to SAE J514, 37-degree two or three piece tube assembly. A system filter shall be furnished in either the suction or return line. Hose reels shall be mounted such that they are within the plan outline of the truck.

3.7.4.1 Hydraulic tubing, hoses, and fittings. All hydraulic tubing, hoses, and fittings shall meet the requirements of the applicable SAE recommended practices J514, J516, J517, J518, J524, and J525 and shall be routed to preclude kinking, chafing, and twisting.

3.8 Structure.

3.8.1. Chassis and frame. The frame and related structures shall be fabricated from structural steel and steel castings at least equal to the size and design of the supplier's standard commercial frame members and shall withstand 300 percent of rated load without permanent deformation. Frame members, bracings, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford

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protection for working parts, and shall withstand, without permanent deformation, stresses induced by capacity loads when tested as specified in 4.3.2.

3.8.2 Uprights and carriage assembly. Uprights shall be the telescopic roller type. Main load rollers shall be the permanently lubricated for life type. The forks and fork carrier shall conform to ANSI B56.1 and MH11.4. The fork carrier shall be free of obstructions to allow the forks to slide across the entire width of the carrier. A cutout shall be provided in the fork carrier so that the lower hook of the fork will fit through the slot and each fork shall be removable with-out removing the load backrest. Fork thickness shall not exceed 1/2 inch at the tips and the end taper shall be not less than 14 inches long; length shall be 36 inches. Forks shall be adjustable to obtain a fork spacing, measured between the forks, of not less than 30 inches.

3.8.3 Wheels. Trucks shall be equipped with one or two drive wheels and two load wheels. Drive wheel(s) shall be of the flat face demountable type, made from steel or cast iron mounted on angular contact ball bearings or tapered roller bearings. Accessible grease fittings shall be provided to facilitate lubrication where sealed bearings are not provided. Wheels, when equipped with new tires, shall clear any part of the truck structure by at least 1/4 inch under all conditions of operation specified herein.

3.8.3.1 Drive wheel tires. The drive wheel tire(s) shall be new and unused, and shall be not more than 12 months old on the date of shipment to the government, and shall be selected from sizes listed in the TRA Yearbook. The tires shall be of the industrial solid rubber or cushion types molded on to a new press-on type steel bond, with nondirectional type thread. Tire loadings shall not exceed the values shown in TRA Yearbook.

3.8.3.2 Load wheels. Trucks shall be equipped with polyurethane load wheels, mounted on angular contact ball bearings or tapered roller bearings. Load wheel ratings shall be sufficient to support maximum loads and operate smoothly without binding.

3.8.4 Steering. The truck shall be steered through the drive wheels by a steering handle equipped with handgrips. The diameter of the handgrips shall be between 1-3/8 and 1-3/4 inches. The truck shall turn in the same direction of the handle when in the forward motion. The forward end of the truck shall be the end containing the control handle. The steering handle shall turn the drive wheel assembly at least 90 degrees in each direction from the neutral position. A safety device, which will cause the truck to move away from the operator in the event the head of the handle should strike the operator, shall be mounted on the head of the steering handle. The steering handle shall return from an operating position to a vertical braking position when released. The force required to hold the steering handle in any operating position shall not exceed 10 pounds.

3.8.5 Brakes. Brakes shall be of the mechanical external contracting or internal expanding or disc type mounted on the traction motor shaft, countershaft, or drive wheel(s). Brake linings shall be of the conventional type, cement bonded to the brake shoes or bolted or riveted to the carrier plate. The brake system shall be interlocked with a relay switch to cause

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disconnection of the electrical circuit from the traction motor when the brakes are applied. The brakes shall automatically be applied when the steering handle is within 10 degrees of the true vertical and horizontal positions. Brakes shall be adjustable to compensate for wear and shall be capable of retaining the truck with rated load on a 5 percent grade. When the high drive control is specified, the service brakes shall be operated by a hand control or a foot control located on the high drive platform. Application of the brakes shall cause disconnection of power to the drive motor. The brakes shall be applied when the operator removes his hand or foot from the controls as applicable.

3.8.6 Truck housing. All functioning components, except steering handle, wheels, battery, and forks shall be enclosed in a housing. The housing shall permit access for inspection, servicing, and maintenance purposes. The height and width of the housing shall be such as to provide unobstructed view of fork tips then the operator is in the operating position.

3.9 Controls. Electric controls shall be heavy duty type. All controls shall be provided with position markings (decals are not acceptable) at the control(s).

3.9.1 Directional speed controls. Directional and speed controls (rotating handles, buttons, levers, or switches) shall be mounted on the steering handle and shall be located for right or left hand operation. Means shall be provided to prevent simultaneous actuation of forward and reverse controls. Controls shall be arranged so that forward movement will cause forward motion of truck and rearward movement will cause rearward motion. Trigger speed controls are not acceptable. If buttons or switches are used, the one located nearest the operator shall, when actuated, cause forward motion of the truck. The controls shall be so constructed that they return to neutral position when released.

3.9.2 Lift and lowering controls. The lifting and lowering control(s) shall be lever actuated type and shall be located for operation by either hand. The lift control shall be so arranged that the forks will remain at any position reached when the control is released. Deactivated controls shall return to the neutral position.

3.9.3 High drive control. When specified, a removable high drive control shall be provided to enable the operator to simultaneously raise or lower the forks and drive forward or reverse in all speeds while standing on the platform of the truck. Steering shall be by means of a vertically mounted steering wheel not to exceed 15 inches in diameter. Speed of the truck shall be limited to low speed with mast raised above 36 inches. The high drive platform shall be 20 inches long and 30 inches wide and shall be provided with safety chains or bars on both sides.

3.9.4 Straddle legs. Straddle legs shall be of sufficient length and width to handle pallets of the size specified.

3.9.5 Hourmeter. An electrically operated hourmeter, which registers the total number of hours of operation of the traction and pump motors shall be provided.

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3.9.6 Horn. Trucks shall be equipped with an electric horn with pushbutton mounted in the steering handle. When the truck is equipped with the high drive control, a horn button shall also be provided on the high drive platform.

3.9.7 Battery discharge indicator. Each truck shall be furnished with a battery discharge indicator containing a lift interrupt. The lift circuit shall be interrupted when the battery has been discharged 80 percent.

3.10 Performance.

3.10.1 Speed. While traveling on a level surface, trucks shall be capable of forward or rearward speeds of not less than 2 mph with rated load and not less than 3 mph without a load.

3.10.2 Lifting speed. Lift speed with rated load (ambient temperature between 65 °F and 85 °F) shall be not less than 15 feet per minute measured over the entire distance from ground level to maximum fork height.

3.10.3 Lowering speed. Lowering speed of the unloaded forks (ambient temperature between 65 °F and 85 °F) shall be from 0 to 25 fpm and shall be infinitely controllable throughout the entire lowering range. The hydraulic control system shall be damped, metered or manually regulated to reduce shock and prevent damage to the trucks when rated load is lowered at maximum speed and abruptly stopped.

3.10.4 Right angle stacking. Trucks, with rated load on the pallet specified (see 3.9.4) shall be capable of right angle stacking in aisles no greater than the pallet length plus 40 inches when tested as specified in 4.3.2.3.

3.10.5 Slope ascension. Trucks shall ascend a 5 percent grade on a concrete surface with and without rated load. Trucks shall accelerate from a dead stop on the slope while carrying the rated load.

3.10.6 Upright height. Maximum collapsed mast height of the unloaded trucks, measured from the ground to the top of the mast shall not exceed 83 inches.

3.10.7 Fork height. Minimum fork height of the loaded trucks, measured from the ground to the top surface of the forks with the mast fully extended shall be 130 inches.

3.10.8 Free lift height. Lift of forks, without exceeding the collapsed mast height, shall be not less than 5 inches with the rated load on the forks.

3.10.9 Underclearance. The truck shall have sufficient underclearance to permit operation from one horizontal plane to another horizontal plane 1 inch lower in depth, without dragging the underside of the truck, and up or down a 10 percent ramp with the rated load on the forks.

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3.10.10 Lift assembly. The lift assembly shall lift and hold the rated load at maximum lift height for 5 minutes with not more than 1 inch vertical drift.

3.10.11 Forks and mast assembly. Forks, mast assembly, and related structures shall show no permanent deformation or failure when tested as specified herein.

3.10.12 Hydraulic system. The hydraulic system shall show no leaks or breakage when the truck is tested as specified in 4.3.2.12 and 4.3.2.13. The pressure relief valve shall open when the pressure equals no more than 150 percent of the maximum pressure needed to lift the rated load to the maximum lift height. The lift system shall lower the rated load at a maximum speed of 80 feet per minute in the event of failure or damage to the hydraulic lines.

3.10.13 Brake. The brake shall stop the truck, traveling at maximum speed with rated load on the forks, within a distance of 3 feet when tested as specified in 4.3.2.9.

3.10.14 Stability. The truck shall conform to the stability requirements of ANSI B56.1 when tested as specified in 4.3.2.10.

3.10.15 Rainproof. When specified (see 6.2), the truck shall be capable of performing all its normal operations after being subject to the rain test specified in 4.3.2.16.

3.10.16 Steering. The force required at the steering handle (with the hand-grip of the steering handle located 40 inches, +1 inch above the floor surface) to turn the drive wheel(s) in either direction when the truck is in a static position with and without the rated load shall not exceed 50 pounds. When the truck is equipped with a removable high drive control, the steering wheel shall turn the truck throughout the 180 degree turning arc, with and without the rated load, with no more than six complete turns of the steering wheel and with no more than 25 pounds of steering effort when tested as specified in 4.3.2.11.

3.11 Lubrication. Accessible means shall be provided for applying and, where necessary, checking the level of lubricant in all parts and components. Oil drain plugs and connections shall provide for instant drainage by using common maintenance shop tools.

3.12 Treatment and painting. The truck shall be cleaned, treated and painted in accordance with MIL-T-704, type A, except that the finish coat shall conform to TT-E-489 eneamel gloss yellow, color number 13538.

3.13 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference emission and susceptibility characteristics of the truck shall conform to MIL-STD-461, class IIID.

3.14 Identification marking. The truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material which shall be in a visible and safe location on the bulkheads or panel of the operator's compartment of the truck.

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The identifying data to be applied to the identification plates shall be as indicated herein and as defined in MIL-STD-130. Identification data shall include National Stock Number (if applicable), contract number, serial number, and US Military Property Number. The truck shall also be equipped with instructions, warning, and caution plates conforming to MIL-P-514, type III, composition C, of type 1, grade A, class 1 material, prominently located and describing any special or important procedure to be followed in operating, loading, and servicing of the truck or its components. All plates shall be securely attached to the truck with screws, bolts, or rivets, and shall be furnished and mounted by the contractor.

3.14.1 Instruction plates. Each truck shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in loading, operating, or servicing the truck. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.15 Workmanship. All workmanship on the truck shall be in accordance with the engineering and production standards of the materials handling industry. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist. Pivot points shall have hardened pins and bushed holes.

3.15.1 Bolted connections. Boltholes may be subpunched or drilled undersize prior to welding and reamed to full size after all welding has been completed, or drilled to full size after all welding has been completed. All holes shall be drilled or reamed through steel templates. Boltholes shall be drilled or reamed straight through from one side of the component.

3.15.2 Welders and welding.

3.15.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS B2.1, Welding Procedure and Performance Qualification.

Certification shall be made available for review by the contracting officer or designated representative upon request. Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be

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responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.15.2.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wirebrushing and other foreign matter. Welds shall transmit stress without permanent deformation or failure when parts connected by the welds are subjected to proof and service loading.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections, evaluations, and demonstrations. Inspections, evaluations, and demonstrations shall be classified as follows:

a. First article evaluation and demonstration (see 4.3).
b. Quality conformance inspection (see 4.4).
C. Inspection of packaging (see 4.6).

4.3 First article evaluation and demonstration. Prior to delivery, the manufacturer shall make one or more trucks available (see 6.2), to be selected at random from trucks being produced from production tooling for first article evaluation and demonstration. The truck and its components shall be compared with this document and the contractor's published literature. Calculations and test data will be used to verify compliance with requirements of this specification. The truck(s) shall be subjected to a demonstration as specified herein. Permanent deformation, overheating, malfunctions, failure to meet performance requirements, failure to complete any test, or leakages of lubricants shall be cause for rejection of the truck(s). If such deficiencies do arise, they shall be corrected and the affected test shall be repeated.

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4.3.1 Schedule of examinations. The following examinations shall be conducted as a part of the first article to assure conformance to section 3 requirements by using the major and minor requirements of 4.4.2.

Major

101.	Dimensions not as specified.
102.	Part missing or not specified type or size.
103.	Component fractured, split, bowed, malformed, loose, not in alinement or not correctly assembled.
104.	Sharp burr, sliver, splinter not ground smooth.
105.	Welding and brazing missing, incomplete, burn holes, cracked, fractured, not fused.
106.	Underclearance not as specified.
107.	UL label missing.
108.	Electromagnetic interference suppression on production trucks not identical with first article test model.
109.	Materials not as specified.
110.	Electrical equipment not as specified, do not function, or are damaged.
111.	Maintainability not as specified.
112.	Lubrication devices missing, not accessible, or not as specified.
113.	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
114.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
115.	Contractor does not have documentation available for identification of material, material finishes or treatments.
116.	Welders and welding certification not as specified.
117.	Identification plates not as specified.
118.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the trucks.

Minor

201.	Finish not dry, free of runs, sags, foreign matter or area of no film, wrong color.
202.	Welding and brazing slag inclusion, scale, or flux deposits not removed, not smooth and uniform.
203.	Threaded fasteners missing, broken, stripped, fractured, or loose.
204.	Identification marking missing, incomplete, not legible.

4.3.2 Tests. Tests shall be conducted as specified herein. All tests shall be made with a fully charged and adequately cycled battery. The procedures outlined herein are intended as guides. Alternate test procedures may be substituted providing the desired results are obtained. However, any deviation from the prescribed procedures shall be subject to prior approval by the contracting officer.

4.3.2.1 Speed. The truck, with capacity load, shall be accelerated on a dry, smooth level surface and timed through an 88-foot interval (88 feet per minute equals 1 mile per hour). Three runs shall be made in both the forward and reverse direction in high speed. This test shall be repeated without

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load. Record data and calculate average speeds. Nonconformance to 3.10.1 shall constitute failure of this test.

4.3.2.2 Lifting speed and lowering speed. Record time in seconds required to raise rated load from floor to maximum fork height using timing device. Measure and record distance in inches from floor to top surface of forks in fully lowered position. Record the difference between maximum fork height and the height of the forks in the lowered position. Record time in seconds to lower unloaded forks from maximum height to fully lowered position. The raising and lowering shall be performed three times. Nonconformance to 3.10.2 or 3.10.3 shall be cause for rejection.

4.3.2.3 Maneuverability. The truck with capacity load uniformly distributed on the size of pallet specified (see 3.9.4), centered laterally on, and square with the forks, shall be operated in 90 degrees intersecting aisles equal to the pallet length plus 40 inches without reversal of direction. This test shall be performed with the center of the pallet 24 inches from the heel of the forks and at a load carrying position of 12 inches above the floor. The truck with the same load shall place and remove the pallet in a stacking space 10 inches wider than the pallet both on the floor and at a height of within 6 inches of the maximum fork height, from an aisle equal to the pallet length plus 40 inches. The truck shall perform this function at least nine out of ten times without reversal of direction and in not more than 60 seconds per removal or replacement cycle. Nonconformance to 3.10.4 shall constitute failure of this test.

4.3.2.4 Slope ascension. The truck with rated load shall be accelerated along a horizontal surface and onto a 5 percent ramp and brought to a dead stop. The travel control shall then be activated and the truck shall accelerate up the ramp. Nonconformance to 3.8.5 or 3.10.5 shall constitute failure of this test.

4.3.2.5 Upright height. The truck shall be placed on a level surface with unloaded forks in the lowest position. Measure in inches the vertical distance from the floor to the uppermost projection of the collapsed mast assembly. Non-conformance to 3.10.6 shall constitute failure of this test.

4.3.2.6 Fork height. Place the truck, with rated load equally distributed on the forks or pallet, on a level concrete surface. Elevate load to the maximum lift height and measure in inches, the distance from the floor to the top face of the forks. Nonconformance to 3.10.7 shall constitute failure of this test.

4.3.2.7 Free lift height. Place the truck on a level surface with rated load equally distributed on the forks or pallet. Elevate forks to maximum height without increasing the specified collapsed mast height. Measure the distance from the floor to the top face of the forks. Nonconformance to 3.10.8 shall constitute failure of this test.

4.3.2.8 Underclearance. Drive the truck from one horizontal plane up a 10 percent ramp to another horizontal plane with the rated load on the forks. Observe whether all parts of the truck clear the surface. Drive the truck

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from one horizontal plane to another horizontal plane 1 inch lower in depth. Non-conformance to 3.10.9 shall constitute failure of this test.

4.3.2.9 Brake. Drive the truck with the rated load on the forks at maximum speed over a level, dry, brushed concrete surface. Apply brakes and measure the distance traveled from the point of application to the point where the truck stops. Nonconformance to 3.10.13 shall constitute failure of this test.

4.3.2.10 Stability. Stability tests shall be performed as specified in ANSI B56.1. Nonconformance to ANSI B56.1 shall constitute failure of this test.

4.3.2.11 Steering. The truck, with and without a rated load on the forks or a pallet, shall be placed on a smooth, dry, level concrete surface with the steering handle grip(s) 40 inches, ±1 inch, above the concrete surface. A tension dynamometer shall be attached at the handgrip. For this test the steering handle holding device shall be made inoperative. A force shall be applied to the tension dynamometer tangent to the arc through which the steering handle moves (horizontal plane). The steering handle shall be made to turn in each direction from the neutral position in an arc of 80 degrees, minimum. Three readings in each direction shall be taken. Record data and calculate average force in each direction. When the high drive control is specified, the steering wheel shall be tested in accordance with test form No. 4 of MIL-STD-268. Nonconformance to 3.10.16 shall constitute failure of this test.

4.3.2.12 Endurance. The truck shall be subjected to 200 cycles of the endurance testing. One cycle shall consist of raising the rated load to 12 inches carrying height, traveling 100 feet forward, raising the load to maximum lift height, lowering it to 12 inches above the floor, traveling 100 feet backward and stopping. At the end of the 200 cycles, the rated load shall be raised to maximum lift height and held for a period of 5 minutes. A gage shall be inserted in the pump pressure line and the pressure required to lift the rated load shall be determined. The cylinder rod shall be fully extended and the relief valve pressure shall be determined. The hydraulic system shall be examined for leakage or breakage. Nonconformance to 3.10.10 or 3.10.12 shall constitute failure of this test.

4.3.2.13 Rate of fall. The forks shall be loaded to rated load and raised to maximum height. The hose connected to the head end of the cylinder shall be disconnected in a manner that simulates catastrophic hose failure. The rate of fall shall be measured with a timing device accurate to ±0.1 second. A rate of fall exceeding 80 feet per minute or evidence of leakage or breakage shall constitute failure of this test.

4.3.2.14 Electromagnetic interference suppression. When specified (see 3.13), the first article model shall be tested to determine compliance with 3.13. The contractor shall furnish the contracting officer a report of the results of the tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.3.2.15 Operation check-out. Each truck as specified in 4.4.3.2 shall be tested to determine that all operating components and systems are in an

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operable condition. The check-out test shall include-traveling (forward and reverse); steering (clockwise and counterclockwise); lifting and lowering rated load; and braking in both horizontal and vertical handle positions. Inability of the truck to perform any function specified shall constitute failure of this test.

4.3.2.16 Rain protection. Test the truck in accordance with test method No. 11. Evidence of moisture in instruments or controls or inability to perform any normal function shall constitute failure of this test.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.3.1. The inspection level shall be level II with an AQL of 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.3.2.1 through 4.3.2.14. The inspection level shall be S-1 with an AQL of 4.0 percent defective for all defects.

4.4.3.2 Individual. Each truck not tested as specified in 4.4.3.1 shall be tested as specified in 4.3.2.15. Failure of the test shall be cause for rejection.

4.5 First article demonstration report. The contractor shall prepare and submit the first article demonstration report to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination and demonstration performed. All deficiencies encountered and actions taken to correct the deficiencies shall be included.

4.6 Inspection of packaging. Inspection of preservation, packing, and marking for level A shall be in accordance with MIL-STD-162. Commercial preservation, packing and marking shall be examined for conformance to ASTM D 3951 and the requirements specified herein.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing, and marking shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Level A preservation, packing, and marking shall be in accordance with the level A requirements for type II of MIL-STD-162.

5.1.2 Commercial. Commercial preservation, packing, and marking shall be in accordance with ASTM D 3951. Additionally, each exterior container shall be marked with the cube and gross weight.

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5.2 Packing. Packing shall be level A or C as specified.

5.2.1 Level A. Each complete truck shall be packed in accordance with the level A requirement of MIL-STD-162, as specified for type II classification.

5.2.2 Level C. Each complete truck shall be packed in a manner which will prevent damage during shipment and subsequent short term covered storage. Components which are especially susceptible to damage or which increase shipping cube shall be removed and either secured to the truck or boxed and secured to the truck.

5.3 Marking. In addition to any special marking required by the contract or order, marking for shipment and storage shall be in accordance with MIL-STD-129 or ASTM D 3951.

6. NOTES

6.1 Intended use. Trucks described herein are intended for moving palletized loads from docks into and around warehouses and stacking these loads. Trucks are also used for stock picking when desired. Trucks are intended for use where floor loading precludes using heavier equipment and are to be used over paved or other hard surfaces.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Size required (see 1.2).
c. When first article test model is not required (see 3.2).
d. Time frame required for submission of first article test model (see 3.2).
e. Number of forklifts required for first article demonstration (see 4.3).
f. When removable high drive control is required (see 3.9.3).
g. Pallet size required (see 3.9.4).
h. When electromagnetic interference suppression is required (see 3.13).
i. When a solid state speed control is required (see 3.7.1.1).
j. Level of preservation and of packing (see 5.1 and 5.2).
k. When a first article demonstration report is required (see 4.5).
l. When rain test is required (see 3.10.15).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first article.

6.4 Safety. Presence of a UL label or submission of a certified test report from a recognized independent testing laboratory may be accepted as evidence that the truck complies with the safety requirements specified in 3.4.1.

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6.5 Material deterioration and control. The truck shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army – ME
Navy - SA

Review activity:
Army - AT
Navy - SH, OS

User activities:
Army - SM
Navy - MC

Preparing activity:
Army - ME

Project 3930-0516

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APPENDIX

FOR

TRUCKS, LIFT, FORK, TIERING, WALKIE

STRADDLE TYPE, ELECTRIC, 2000- and 2,500 POUNDS

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether forklifts procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

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TEST METHOD NO. 1

RAIN TEST

1. Test apparatus:

   a. Timing device.
   b. Measuring device (U.S. Weather Bureau type preferred).
   c. Spray fixture.

2. Test procedure:

   a. Subject the truck to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the truck at an angle of 45 degrees ±5 degrees with a force equivalent to a 30 mph wind. The simulated rainfall shall dispense uniformly over the surface of the truck.
   b. Expose the top and left side of the truck to the rain described above for a period of 5 minutes.
   c. Repeat "b" with top and front exposed.
   d. Repeat "b" with top and front side exposed.
   e. Repeat "b" with top and rear exposed.
   f. Examine all instruments and electric controls for evidence of moisture.
   h. Demonstrate operation of the truck.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52698A	2. DOCUMENT DATE (YYMMDD) 860314

3. DOCUMENT TITLE
TRUCKS, LIFT, FORK, TIERING, WALKIE, STRADDLE TYPE, ELECTRIC, 2,000- AND 2,500-POUNDS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-E-ELE/21 (4EELD)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-6516 (2) DSN 786-6516
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-E-ELE/21 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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